J. Function Block Reference

Appendix J of the EFC Gases & Advanced Materials Controls Standard, Revision I, October 14, 2026. Standalone copy; see the full document for the other chapters and appendices.

This appendix documents the commonly used Derived Function Blocks (DFBs) in EFC's Control Expert projects, taken from the Control Expert data editor. Section 10.8 of the Controls Standard explains how DFBs are designed and managed. For each block, the pin table lists its inputs, outputs, and input/outputs; where a pin is a data type, its members are listed in Appendix H of the Controls Standard and are not repeated here. Internal timers and working variables are omitted. Descriptions marked *TBD* and names marked *[check]* still need to be confirmed.

J.1 DFB_Analog_Input

Scales an X80 analog input channel into a process value and passes the channel's health and diagnostic bits to the process value structure. Pairs with `DDT_Process_Value`.

Table J.1: DFB_Analog_Input Pins

Pin	Name	Direction	Type	Description
1	`Input`	Input	T_U_ANA… *[check]* (X80 analog channel device DDT)	The analog input channel from the I/O module; members below.
1	`Raw_Value`	Output	INT	Unscaled channel value.
2	`Eng_Value`	Output	REAL	Scaled value in engineering units.
3	`PV`	Input/output	DDT_Process_Value	Process value structure, including its LL/L/H/HH alarms (`DDT_AI_Alarm`), `HW_Fault`, `Value_OOR`, `Underrange`, and `Overrange`.

Public variables: `Overrange` and `Underrange` (BOOL).

Table J.2: X80 Analog Channel Members Used by DFB_Analog_Input

Member	Type	Description (Schneider)
`FCT_TYPE`	WORD	Function type

Member	Type	Description (Schneider)
CH_HEALTH	BOOL	Channel health
CH_WARNING	BOOL	Channel warning
ANA.VALUE	INT	Analog input value
ANA.FORCED_VALUE	INT	Forced value
ANA.FORCE_CMD	BOOL	Force command
ANA.FORCED_STATE	BOOL	Forced state
ANA.TRUE_VALUE	INT	Physical value
MEASURE_STS	INT	Measurement status
CH_ALIGNED	BOOL	Aligned channel
LOWER_LIMIT	BOOL	Measurement within lower limit
UPPER_LIMIT	BOOL	Measurement within upper limit
INT_OFFSET_ERROR	BOOL	Internal offset error
INT_REF_ERROR	BOOL	Internal reference error
POWER_SUP_ERROR	BOOL	Power supply error
SPI_COM_ERROR	BOOL	SPI communication error

J.2 DFB_Process_Value

Evaluates the four analog alarms (LL, L, H, HH) of a process value, applying each alarm's on-delay, deadband, enable, and inhibit settings. Pairs with DDT_Process_Value and DDT_AI_Alarm.

Table J.3: DFB_Process_Value Pins

Pin	Name	Direction	Type	Description
1	PV	Input/output	DDT_Process_Value	Process value whose LL/L/H/HH alarms are evaluated.
2	Reset	Input	BOOL	Resets latched alarms *TBD*.

No outputs or public variables. Internally, one on-delay timer per alarm (LL_TMR, L_TMR, H_TMR, HH_TMR) uses each alarm's DLY setting.

J.3 DFB_Device

Controls a motor or other on/off device: Off/Manual/Auto/Lockout modes, run command, feedback, runtime tracking, and stall detection. Pairs with DDT_Device.

Table J.4: DFB_Device Pins

Pin	Name	Direction	Type	Description
1	DEV	Input/output	DDT_Device	Device data structure (requests, permissives, command, feedback, alarms, runtime).

Table J.4 continued

Pin	Name	Direction	Type	Description
?	Inputs *[check]*	Input	—	The block's input pins were not visible in the reference screenshots; add them here.

Public variables: `HMI_CMD` (BOOL), `Mode` (INT; Off, Auto, and Manual values from `DDT_Mode`), `Runtime` (REAL), `Reset_Runtime` (BOOL), and `Motor_Stalled` (BOOL). Code sections: `Runtime_Tracking` (structured text) and `Commands`.

J.4 DFB_Valve

Controls an on/off valve with optional open/closed limit switch feedback: modes, open command, interlock and E-stop handling, feedback checking, and cycle counting. Pairs with `DDT_Valve`. Written in ladder diagram (one `Commands` section).

Table J.5: DFB_Valve Pins

Pin	Name	Direction	Type	Description
1	`Valve`	Input/output	DDT_Valve	Valve data structure.
2	`Valve_Output`	Output	BOOL	Output to the valve solenoid.
3	`Interlock`	Input	BOOL	Interlock; when active the valve goes to its safe state.
4	`Estop`	Input	BOOL	E-stop status; when active the valve goes to its safe state.
5	`Sys_Auto`	Input	BOOL	System (unit) is in Auto mode.
6	`Sys_Manual`	Input	BOOL	System (unit) is in Manual mode.
7	`Sys_Off`	Input	BOOL	System (unit) is in Off mode.

No public variables. Internally, on- and off-delay timers handle feedback checking.

J.5 DFB_Station

Runs one cylinder fill station: selection, venting, evacuation, pressurizing, sampling, filling, and disconnecting, with its scale, sequence, pressure transmitters, and three valves. Pairs with `DDT_Station`. Code sections: `DataMapping` and `Monitoring` (structured text).

Table J.6: DFB_Station Pins

Pin	Name	Direction	Type	Description
1	`Station`	Input/output	DDT_Station	Station data structure.
2	`Scale`	Input/output	DDT_Scale	Station scale.
3	`Seq`	Input/output	DDT_Seq	Station fill sequence.
4	`PT`	Input/output	DDT_Process_Value	Station pressure transmitter.

Table J.6 continued

Pin	Name	Direction	Type	Description
5	`InletValve`	Input/output	DDT_Valve	Inlet valve.
6	`SampleValve`	Input/output	DDT_Valve	Sample valve.
7	`VacuumValve`	Input/output	DDT_Valve	Vacuum valve.
8	`Estop`	Input	BOOL	E-stop status.
9	`Upstream_PT`	Input	DDT_Process_Value	Upstream (supply) pressure transmitter.

Public variables: `ONS_Auto` and `ONS_Manual` (BOOL; one-shots on a change to Auto or Manual *TBD*), and the station status values below, which are written to `DDT_Station.StatusEnum`.

Table J.7: DFB_Station Status Values (StatusEnum)

Status	Value
`Disabled`	0
`ReadyToStart`	10
`ReadyToFill`	20
`Filling`	30
`Complete`	40
`Failed`	50
`Venting`	60
`Vacuuming`	70